Question # 6 of 10 (Start time: 08:38:25 AM, 25 January 2024)

The condition $|f(t)e^{-kt}| \leq M$ ensures that the function is bounded by a constant value. What is the significance of this boundedness?

Select the correct option

○	It ensures the periodicity of the function.
○	It guarantees the continuity of the function.
◉	It ensures the existence of the Laplace transform. Correct 💯
○	It guarantees the differentiability of the function.

Click to Save Answer &

Mth 647 Quizz No: 3

Module - 81 from Module - 89

Edit: Khalil Ahmad (M.sc Math)

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Question # 3 of 10 (Start time: 08:37:02 AM, 25 January 2024)

What does the expression $s\mathcal{L}(f(t)) - f(0)$ represent in the context of Laplace transforms?

Select the correct option

○	The Laplace transform of the second derivative of $f(t)$
○	The Laplace transform of the original function $f(t)$
○	The Laplace transform of the integral of $f(t)$
○	The Laplace transform of the derivative of $f(t)$ Correct 💯

Click to Save Answer &

Mth 647 Quizz No: 3

Module - 81 from Module - 89

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The Laplace transform of $f(t) = t^3$ is given by:

Select the correct option

○	$\frac{3}{s^4}$.
○	$\frac{4}{s^3}$.
○	$\frac{3s}{s^4}$.
◉	$\frac{6}{s^4}$.

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Correct 💯

Question # 10 of 10 (Start time: 12:31:53 AM, 25 January 2024)

What is the Laplace transform of $e^{at} f(t)$ denoted as?

Select the correct option

- $F(s+a)$ Regards: 🌹 Khalil Ahmad (M.sc Math) 🌹 PAPR01 - Virtual University Muzaffargarh Road Campus, Ali Pur - First Floor UBL
- $F(s)$
- $F(s \times a)$
- (●) $F(s-a)$ Correct 💯

Within each interval, a piecewise continuous function must be:

Select the correct option

Reload Math Equations

- ○ Differentiable
- ○ Monotonic
- ◉ Continuous
- ○ Discontinuous

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Correct 💯

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If $f(t) = t$, then the Laplace transform of $f'(t)$ is:

Select the correct option

Reload Math Equations

- (●) $\frac{1}{s}$
- ( ) s
- ( ) $1 + \frac{1}{s}$
- ( ) $1 - \frac{1}{s}$

Correct 💯

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For the function $f(t)$ having the Laplace transform $\mathcal{L}(f(t))$,the relation between $F(s)$ and $F(s-a)$ is given by;

Select the correct option

Reload Math Equations

- (●) s-Shifting property

Correct 💯

- ( ) Periodic property

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- ( ) Convolution property
- ( ) Symmetric property

Show Answer

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If $f(t) = e^t$, then the Laplace transform of $f'(t)$ is:

Select the correct option

Reload Math Equations

- $\frac{s^2}{s-1}$
- None of other choices
- $\frac{s}{s-1}$
- $\frac{s}{s+1}$

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Correct 💯

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sec(s)

MTH647 - Methods in Mathematical Physics (Quiz - 03)

Quiz Start Time: 03:39 AM, 24 January 2024

Question # 5 of 10 (Start time: 03:41:07 AM, 24 January 2024)

Total Marks: 1

The expression $\frac{(n+1)!}{s^{n+2}}$ is obtained by substituting the Laplace transform of t^n in the inductive step(last step). What does this expression represent?

Select the correct option

Reload Math Equations

- Laplace transform of t^n
- Laplace transform of t^{n+1}
- Laplace transform of t^{2n}
- Laplace transform of t^{n+2}

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Correct 💯

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Laplace transform of t^n is given by: $\mathcal{L}\{t^n\} = ?$

Select the correct option

Reload Math Equations

- $\frac{(n+1)!}{s^n}$
- $\frac{n!}{s^{n+1}}$
- $\frac{(n-1)!}{s^n}$
- $\frac{n!}{s^n}$

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Correct 💯

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What is the base case(initial step) for the proof by mathematical induction for the Laplace transform of t^n?

Select the correct option

Reload Math Equations

- ○ $n = 2$
- ○ $n = -1$
- ○ $n = 0$
- ◉ $n = 1$

Correct 💯

Show Answer

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sec(s)

MTH647 - Methods in Mathematical Physics (Quiz - 03)

Quiz Start Time: 03:39 AM, 24 January 2024

Question # 8 of 10 (Start time: 03:44:13 AM, 24 January 2024)

Total Marks: 1

The Gamma function is a generalization of the:

Select the correct option

Reload Math Equations

- (●) Factorial function
- ( ) Exponential function
- ( ) Sine function

Correct 💯

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The condition $|f(t)e^{-kt}| \leq M$ ensures that the function is bounded for all $t \geq 0$. What does M represent in this condition?

Select the correct option

Reload Math Equations

- Minimum value of the function
- Maximum value of the function
- Upper bound on the function
- Lower bound on the function

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Correct 💯

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sec(s)

MTH647 - Methods in Mathematical Physics (Quiz - 03) **Quiz Start Time: 03:39 AM, 24 January 2024**

Question # 10 of 10 (Start time: 03:45:15 AM, 24 January 2024) **Total Marks: 1**

The Laplace transform of $f(t) = t^{\alpha}$, where 'α' is any constant, can be found using:

Select the correct option

Reload Math Equations

- Partial fraction decomposition
- Chain rule.
- Integration by parts
- L'Hôpital's rule

Correct 💯

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MTH647 - Methods in Mathematical Physics (Quiz - 03)

Quiz Start Time: 01:32 AM, 24 January 2024

Question # 1 of 10 (Start time: 01:32:11 AM, 24 January 2024)

Total Marks: 1

If $f(t) = \sin t$, then the Laplace transform of $f'(t)$ is:

Select the correct option

Reload Math Equations

- $\mathcal{L}(f\prime(t)) = \frac{s^2}{s^2+1}$
- $\mathcal{L}(f\prime(t)) = \frac{-s}{s^2+1}$
- $\mathcal{L}(f\prime(t)) = \frac{-s^2}{s^2+1}$
- $\mathcal{L}(f\prime(t)) = \frac{s}{s^2+1}$

Show Answer

Remember in your precious prayers

Correct 💯

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MTH647 - Methods in Mathematical Physics (Quiz - 03)

Quiz Start Time: 01:32 AM, 24 January 2024

Question # 6 of 10 (Start time: 01:39:07 AM, 24 January 2024)

Total Marks: 1

Where can discontinuities occur in a piecewise continuous function?

Select the correct option

Reload Math Equations

Show Answer

- Only at the critical points of the function.
- Discontinuities can occur at the endpoints of each interval and within the intervals.
- Only at the endpoints of each interval.
- Only in the middle of each interval.

Correct 💯

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MTH647 - Methods in Mathematical Physics (Quiz - 03)

Quiz Start Time: 01:32 AM, 24 January 2024

Question # 7 of 10 (Start time: 01:40:51 AM, 24 January 2024)

Total Marks: 1

What is a piecewise continuous function?

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Select the correct option

Reload Math Equations

Show Answer

- A function that has a continuous derivative.
- A function that is continuous everywhere. ✓
- A function that has no discontinuities.
- A function that is defined by different formulas on different intervals.

correct 100

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MTH647 - Methods in Mathematical Physics (Quiz - 03)

Quiz Start Time: 01:32 AM, 24 January 2024

Question # 9 of 10 (Start time: 01:44:39 AM, 24 January 2024)

Total Marks: 1

What is the Laplace transform of the derivative of a function $f(t)$ denoted by $f\prime(t)$?

Select the correct option

Reload Math Equations

- (●) $s\mathcal{L}\{f(t)\} - f(0)$
- ( ) $\mathcal{L}\{f(t)\}$
- ( ) $s\mathcal{L}\{f(t)\}$
- ( ) $f(0) - s\mathcal{L}\{f(t)\}$

Correct 💯

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